

Detailed Performance Summary For Qx_100to20_dTe_const_Tc_41to20C

Project: 30XW_China
Prepared By: Mikhail Terekhov

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AquaForce Greenspeed Water Cooled Screw Chiller



Unit Information

Tag Name: **Qx_100to20_dTe_const_Tc_41to20C**
Model Number: **30XW-VB459**
Condenser Type: **Water Cooled**
Nameplate Voltage: **400-3-50** V-Ph-Hz
Quantity: **3**
Manufacturing Source: **Shanghai, China**
Chiller Type: **Cooling**
Operating Mode: **Cooling**
Refrigerant: **R-515B**
Minimum Capacity: **20.00** %
Shipping Weight: **9421** kg
Operating Weight: **9327** kg
Unit Length: **4524** mm
Unit Width: **1745** mm
Unit Height: **2632** mm

Performance Information

Cooling Capacity: **1 016** kW
Heating Capacity: **1 268** kW
Total Unit Power: **285.8** kW
Cooling Efficiency (COPR): **3.554** kW/kW
Heating Efficiency (COPH): **4.439** kW/kW

Acoustics Information

A-Weighted Sound Pressure Level: **81** dBA

Evaporator Information

Fluid Type: **Fresh Water**
Fouling Factor: **0.0180** (sqm-K)/kW
Leaving Temperature: **6.00** °C
Entering Temperature: **13.00** °C
Fluid Flow: **34.59** L/s
Pressure Drop: **15.2** kPa

Condenser Information

Fluid Type: **Propylene Glycol**
Brine Concentration: **45.0** %
Fouling Factor: **0.0440** (sqm-K)/kW
Leaving Temperature: **49.00** °C
Entering Temperature: **41.00** °C
Fluid Flow: **41.72** L/s
Pressure Drop: **24.1** kPa

Electrical Information

Unit Voltage: **400-3-50** V-Ph-Hz

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Standby Power:.....**0.05000** kW
Minimum Voltage:.....**360** Volts
Maximum Voltage:.....**440** Volts
Power Factor:.....**0.93**

Amps (Un)	Electrical Circuit 1	Electrical Circuit 2
Max Unit Current Draw (RLA)	464.0	---
Nominal Start Up Current (ICF)	294.0	---
Nominal Unit Current Draw (A)	294.0	---

Accessories and Installed Options

Opt. 282 EMC C2 Classification
Opt. 001A Integrated VFD

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Load Line

Unit Performance	**	***								
User-specified Cooling Capacity, kW		1000.0	900.0	800.0	700.0	600.0	500.0	400.0	300.0	200.0
Percent of User-specified Cooling Capacity, %	102	100.00	90.00	80.00	70.00	60.00	50.00	40.00	30.00	20.00
Percent Full Load Cooling Capacity, %	100.0	98.46	88.62	78.77	68.92	59.08	49.23	39.38	29.54	19.69
Percent of Full Load Power, %	100.0	98.16	79.57	66.41	50.14	40.04	31.00	21.82	13.24	7.572
Unloading Sequence	Default	Default	Default	Default	Default	Default	Default	Default	Default	Default
Cooling Capacity, kW	1 016	1 000.	900.0	800.0	700.0	600.0	500.0	400.0	300.0	200.0
Heating Capacity, kW	1 268	1 248	1 101	967.1	825.9	700.3	577.3	454.0	332.5	219.4
Total Unit Power, kW	285.8	280.5	227.4	189.8	143.3	114.4	88.58	62.36	37.84	21.64
Nominal Unit Current Draw (RLA)	437.17	429.8	351.3	289.8	219.4	177.5	138.6	97.7	59.3	33.9
Cooling Efficiency kW/kW	3.554	3.565	3.958	4.215	4.885	5.243	5.645	6.414	7.928	9.242
Heating Efficiency, kW/kW	4.439	4.449	4.840	5.096	5.764	6.120	6.517	7.280	8.787	10.14
A-Weighted Sound Pressure Level, dBA	81	81	80	78	77	78	74	73	73	73
Evaporator Data										
Fluid Entering Temperature, °C	13.00	13.00	13.00	13.00	13.00	12.86	11.72	10.57	9.43	8.29
Fluid Leaving Temperature, °C	6.00	6.00	6.00	6.00	6.00	6.00	6.00	6.00	6.00	6.00
Fluid Flow Rate, L/s	34.59	33.99	30.59	27.19	23.79	20.81	20.80	20.80	20.80	20.80
Pressure Drop, kPa	15.2	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0
Fouling Factor, (sqm-K)/kW	0.0180	0.0180	0.0180	0.0180	0.0180	0.0180	0.0180	0.0180	0.0180	0.0180
Condenser Data										
Fluid Entering Temperature, °C	41.00	41.00	40.00	39.00	35.00	32.00	29.00	26.00	23.00	20.00
Fluid Leaving Temperature, °C	49.00	49.00	47.09	45.23	40.33	36.52	32.73	28.94	25.15	21.42
Fluid Flow Rate, L/s	41.72	40.91	40.94	40.94	40.91	40.87	40.82	40.80	40.81	40.80
Pressure Drop, kPa	24.1	23.2	23.4	23.6	24.3	24.9	25.6	26.3	27.1	27.9
Fouling Factor, (sqm-K)/kW	0.0440	0.0440	0.0440	0.0440	0.0440	0.0440	0.0440	0.0440	0.0440	0.0440

Note:

(**): Column displaying effective unit full load performances

(***): Column displays unit performance at the 'required capacity' and design conditions specified

Note: Performance is at nominal voltage. Chiller may unload if job site voltage is lower than 400 volts.

Sound pressure level was determined in accordance with AHRI Standard 575. Sound pressure level is the free field sound data (the unit is placed in free field over a reflecting plane).

Outside the scope of AHRI Water-Cooled Water-Chilling and Heat Pump Water-Heating Packages Certification Program or not optionally certified, but is rated in accordance with AHRI Standard 550/590 (I-P) and AHRI Standard 551/591 (SI).